

FOORT: Flexible Object-Oriented Ray Tracing

Daniel R. Mayerson¹, Fabio Bacchini^{2,3}, Seppe J. Staelens⁴

¹ Institute for Theoretical Physics, KU Leuven, Celestijnenlaan 200D, B-3001 Leuven, Belgium

² Centre for mathematical Plasma Astrophysics, KU Leuven, Celestijnenlaan 200B, B-3001 Leuven, Belgium

³ Royal Belgian Institute for Space Aeronomy, Solar-Terrestrial Centre of Excellence, Ringlaan 3, 1180 Uccle, Belgium

⁴ DAMTP, Centre for Mathematical Sciences, University of Cambridge, Wilberforce Road, Cambridge CB3 0WA, United Kingdom

daniel.mayerson@kuleuven.be, fabio.bacchini@kuleuven.be, ss3033@cam.ac.uk

Abstract

We present FOORT, a C++ code for Flexible Object-Oriented Ray Tracing that is designed with modularity and adaptability in mind. FOORT has multiple metrics already implemented, ranging from Kerr to fuzzballs and boson stars. New metrics can easily be added, whether they are analytical or numerical. This paper presents the main code features and demonstrates its capabilities by presenting several tests and results. The code provides good scaling with an increasing number of threads, making it well-suited to be executed on high performance computing systems too. The code has been made public in a GitHub repository.

Contents

1	Introduction	1
2	Basic Ray Tracing Code Features	1
2.1	Geodesic initialization	1
2.2	Integration schemes	1
2.3	Tests & performance	2
3	Specific FOORT Code Features	4
3.1	Output & analysis	5
3.2	Adaptive mesh integration	6
3.3	Emission models and interferometry	7
4	Demonstrations & Applications	9
A	Implemented metrics	10
A.1	Black holes	11
A.2	Fuzzball	13
A.3	Solitonic boson star	13

B	Expressions for the fluid velocity model	14
B.1	Circular velocity outside the ISCO	14
B.2	Circular velocity inside the ISCO	14
B.3	Radially infalling velocity	14

1 Introduction

(FB: maybe a recap of what's been going on in the community, flex on us initiating the thing, etc?)

2 Basic Ray Tracing Code Features

The geodesic equations are cast in a first-order system;

$$\frac{dx^\mu}{d\tau} = u^\mu, \quad (2.1)$$

$$\frac{du^\mu}{d\tau} = -\Gamma_{\lambda\sigma}^\mu u^\lambda u^\sigma, \quad (2.2)$$

where τ is an affine parameter and $\Gamma_{\lambda\sigma}^\mu$ are the usual Christoffel symbols.

2.1 Geodesic initialization

Vielbein construction.

2.2 Integration schemes

FOORT implements standard integration schemes for numerically solving the geodesic equations (2.1, 2.2). At the moment, the code supports the explicit Runge-Kutta scheme of order 4 (RK4) and the second-order explicit Verlet algorithm. The latter is computationally cheaper than RK4 and possesses better geometric properties for orbit integration (see e.g. [1]) *(SS: Hier stond ook Hairer 2005 - weet niet welke dit is)*. By default, RK4 is selected.

The calculation of the Christoffel symbols requires spacetime derivatives of the metric, which are calculated using second-order finite differencing. *(SS: Would it make sense to make this fourth-order accurate for the RK4 scheme?) (FB: need to say something about derivatives? — yes)*

Given an ODE of the form

$$\frac{dy}{dt} = f(t, y(t)) \quad (2.3)$$

and the value y^n at time t^n , the RK4 scheme advances the variable $y(t)$ over a time step Δt according to

$$\begin{aligned} k_1 &= f(t^n, y^n), \\ k_2 &= f(t^n + \Delta t/2, y^n + k_1 \cdot \Delta t/2), \\ k_3 &= f(t^n + \Delta t/2, y^n + k_2 \cdot \Delta t/2), \\ k_4 &= f(t^n + \Delta t, y^n + k_3 \cdot \Delta t), \end{aligned}$$

$$y^{n+1} = y^n + \frac{\Delta t}{2} (k_1 + 2k_2 + 2k_3 + k_4).$$

This scheme is fourth-order accurate and is applied to both (2.1) and (2.2).

The Verlet scheme advances the geodesic equations as follows [4]:

$$x_{n+1}^\mu = x_n^\mu + u_n^\mu \cdot \Delta\tau + \frac{(\Delta\tau)^2}{2} \left(\frac{du^\mu}{d\tau} \right)_n, \quad (2.4)$$

$$u_{n+1,p}^\mu = u_n^\mu + \left(\frac{du^\mu}{d\tau} \right)_n \Delta\tau, \quad (2.5)$$

$$\left(\frac{du^\mu}{d\tau} \right)_{n+1} = -\Gamma_{\lambda\sigma}^\mu(x_{n+1}^\mu) u_{n+1,p}^\lambda u_{n+1,p}^\sigma, \quad (2.6)$$

$$u_{n+1}^\mu = u_n^\mu + \frac{\Delta\tau}{2} \left[\left(\frac{du^\mu}{d\tau} \right)_n + \left(\frac{du^\mu}{d\tau} \right)_{n+1} \right]. \quad (2.7)$$

Steps (2.6) and (2.7) can be repeated until the change in the velocity vector is below some predefined tolerance (default is a relative change of 10^{-3}), which will improve the accuracy. The advantage of the Verlet scheme is that the Christoffel symbols only need to be evaluated twice, at x_n^μ and x_{n+1}^μ , irrespective of this final iteration. *(SS: I implemented this change in the code.)*

Furthermore, FOORT also makes use of the adaptive step-size routine introduced in [4, 13] and used in other ray tracing codes like [2]. The step-size $\Delta\tau$ is determined by

$$\Delta\tau = \epsilon \left[|\Delta\tau_1|^{-1} + |\Delta\tau_2|^{-1} + |\Delta\tau_3|^{-1} \right]^{-1}, \quad (2.8)$$

where

$$\Delta\tau_1 = (|u^r| + \delta)^{-1} \quad (2.9)$$

$$\Delta\tau_2 = \min(\theta, \pi - \theta) \cdot (|u^\theta| + \delta)^{-1} \quad (2.10)$$

$$\Delta\tau_3 = (|u^\phi| + \delta)^{-1}. \quad (2.11)$$

In this setup, δ is a small positive parameter - set to 10^{-20} - that prevents division by 0, whereas ϵ determines the overall scale of the step size. The code also has a parameter setting a minimum step-size, defaulted to 10^{-12} . The above routine improves accuracy and efficiency in regions where the numerical integration becomes challenging, like in the vicinity of the event horizon and the poles.

2.3 Tests & performance

To test the code's performance on parallel architectures, we carry out a standard geodesic calculation for a Kerr BH with spin $a = 0.99$. We employ a uniform mesh with 16384^2 pixels, and set the camera parameters as in XXXXXXXX. Numerical-integration parameters are as in XXXXXXXX. This setting provides a scenario that is relatively unbalanced for parallel calculations: the BH's shadows covers a fraction XXXXXXXX of the viewscreen, implying that many geodesics (all those that fall into the BH) will be integrated for a much shorter time (i.e. until they reach the horizon) than those that hover just around the shadow's edge. Geodesics passing sufficiently far from the BH will also take a much shorter time to be integrated. In essence, a handful of light rays that are integrated for a long time are spread across a few threads, and are therefore driving the overall code's performance.

The results are shown in Fig. 1, where we report the code's speedup from the baseline (i.e. the runtime on 1 OMP thread) up to 128 threads. We observe that the code performs fairly well throughout the scaling, in spite of the unbalance brought about by this choice of metric. At 128 threads, the measured speedup deviates from the ideal scaling by a factor ~ 2 .

(FB: scaling with a more “uniform” (less unbalanced) test — schwarz, scaling with flat space in the same figure)

Ultracompact spacetimes contain bound null geodesics that neither escape nor fall into a potential horizon. These bound geodesics lead to photon rings; thin, nearly circular features

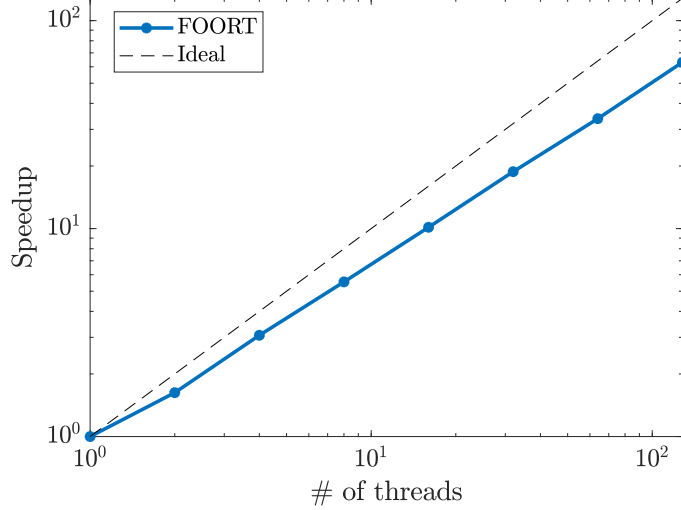


Figure 1: Scaling test for **FOORT** employing a standard Kerr BH with spin $a = 0.99$ and a uniform grid with 16384^2 pixels.

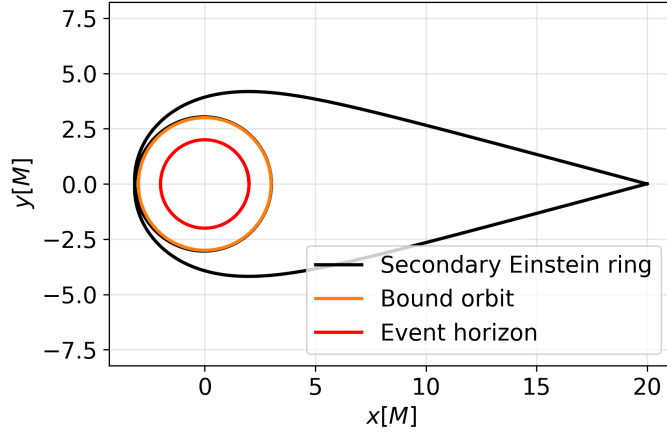


Figure 2: Visualization of the null geodesics in Schwarzschild spacetime relevant for testing the code’s performance. Black: a geodesic creating a secondary Einstein ring at $r = 20M$. Orange: the bound orbit at $r = 3M$. The red circle represents the event horizon at $r = 2M$.

that are expected to contribute significantly to images of black holes and dominate on long interferometric baselines [8] - see e.g. [10] for a recent review. The photon rings are caused by nearly bound geodesics, that approximate the bound ones: the radial coordinate of the nearly bound geodesic exhibits a characteristic exponential divergence away from the radius r_b of the bound geodesic, determined by the Lyapunov exponent γ_L . We have used **FOORT** to integrate the bound circular geodesic at $r = 3M, \theta = \frac{\pi}{2}$ in Schwarzschild spacetime. Due to discretization errors, the geodesic quickly slightly deviates from $r = 3M$, therefore becoming nearly bound. Figure 3 shows the value of the radial coordinate as function of the number of half-orbits (corresponding to ϕ changing by π), showing that the geodesic indeed exponentially diverges away from $r = 3M$, at the correct rate as set by the Lyapunov exponent for Schwarzschild spacetime $\gamma_L = \pi$. This test demonstrates that **FOORT** is able to reproduce the characteristic behaviour of nearly bound orbits.

As a second test, we use **FOORT** to integrate the black geodesic shown in Fig. 2. This geodesic starts at an observer located at $(r = 20M, \theta = \frac{\pi}{2}, \phi = 0)$, to which it returns after orbiting the black hole twice, giving rise to a secondary Einstein ring. The coordinate time

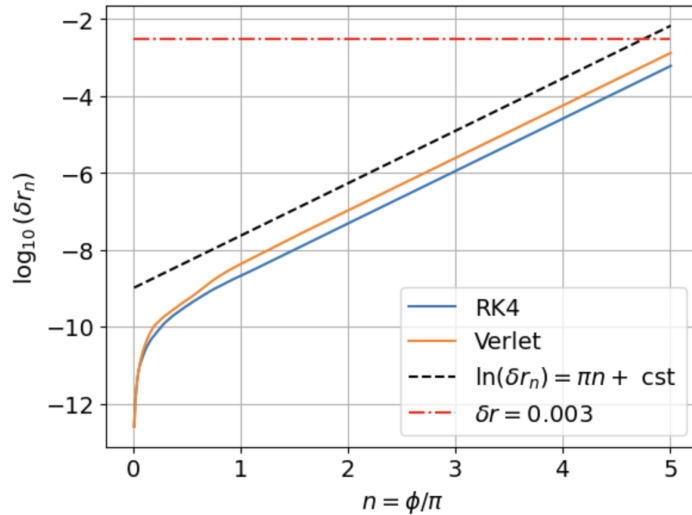


Figure 3: Deviation of the radial coordinate of a geodesic, initialized at $x^\mu = (0, 3M, \pi/2, 0)$ with velocity vector tangent to ∂_t and ∂_ϕ . Theoretically, the geodesic should remain at this fixed radius, but numerical errors quickly perturb it, after which the growth in the deviation follows the characteristic exponential divergence as function of the number of half-orbits n , set by the Lyapunov exponent $\gamma_L = \pi$. Both integration schemes perform comparatively well, with a fractional error below 0.1% for the first 5 half-orbits.

elapsed for this geodesic is calculated to be $\Delta t = 100.792M$ - see [1, 12]. The initial velocity vector of the geodesic is given by

$$\mathbf{v} = \left(\frac{1}{\sqrt{-g_{tt}(r_0)}}, -\frac{\cos \xi}{\sqrt{g_{rr}(r_0)}}, 0, \frac{\sin \xi}{r_0} \right), \quad (2.12)$$

where $\xi = 0.24904964$. We stop the integration of the geodesic once it passes $r = 20M$ again, using the boundary sphere termination (see Sec. 3.1), and save the final position $\mathbf{x}_f$ of the geodesic. We then compare the difference between this position and the expected values using the metric

$$\Delta l = \sqrt{(t_f - t_f^*)^2 + r_f^2 + (r_f^*)^2 - 2r_f r_f^* \cos(\phi_f - \phi_f^*)}, \quad (2.13)$$

where the $(t_f^*, r_f^*, \phi_f^*) = (100.792, 20M, 0)$ are the analytic values mentioned before. Figure 4 shows the values of this metric as function of the number of integration step N . The number of steps is not directly controlled, but rather a consequence of the value of the parameter ϵ in the adaptive step size routine (2.8). While the RK4 integrator clearly outperforms the Verlet integrator in the regime considered, the results for the latter display the expected second-order convergence. Normally, RK4 is fourth-order accurate, but the value of the metric seems to be decreasing very slowly as the number of integration steps increases. This is likely due to the accumulation of errors around the time that the geodesic approaches the bound circular orbit, as the RK4 scheme does not preserve the values of the conserved quantities like the energy – similar to what [1] found. It should also be noted that a more accurate test could be performed by interpolating the last few points on the geodesic to estimate t, ϕ when $r = 20M$, rather than simply taking the last point on the geodesic which will have a value for r close to, but not exactly equal to $20M$.

3 Specific FOORT Code Features

FOORT is designed with flexibility in mind: in particular, it is designed to be as straightforward as possible to add new metrics, diagnostics, and terminations. Furthermore, the code is made

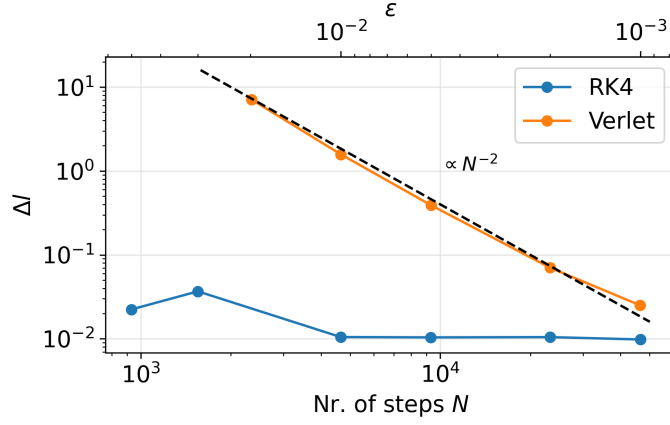


Figure 4: Distance metric (2.13) as function of the number of integration steps for the black geodesic in Fig. 2. The number of integration steps is controlled by the parameter ϵ in (2.8). The Verlet method exhibits its second-order convergence, whereas the RK4 remains fairly constant at a lower value.

efficient through the use of an adaptive mesh, enabling the image to obtain high resolution in the regions of interest.

3.1 Output & analysis

The geodesic integrator will keep on computing trajectories until predetermined termination conditions are met. Depending on the metric of interest, different terminations for the geodesic integrator can be included:

- **Boundary sphere:** The geodesic integrator will stop once the geodesic arrives at a (fictitious) boundary sphere with a tunable radius. This should be large enough such that geodesics reaching this boundary sphere can really be considered as 'escaped to infinity'.
- **Time out:** One can set a limit on the number of integration steps that are taken per geodesic. This is particularly useful for geodesics that approach bound orbits or coordinate singularities.
- **Event horizon:** For certain metrics with an event horizon (currently only horizons at a fixed value of the radial coordinate), a parameter ϵ_H can be set such that the geodesic integration is terminated once the geodesic reaches $r_H(1 + \epsilon_H)$.
- **θ -singularity:** Metrics where one of the coordinates is the polar angle θ typically become singular at the poles. To avoid numerical instability, geodesic integration is terminated once the geodesic position approaches the poles within a tunable $\Delta\theta < \epsilon_\theta$.

Additionally, geodesic integration can also be terminated when a NaN is encountered, and general singularities of any codimension can be added.

At the moment, FOORT allows the user to track the following diagnostics:

- **Geodesic position:** When switched on, FOORT will keep track of the entire geodesic trajectory and output it to a file. This is useful for visualization purposes.
- **Four-color screen:** This diagnostic assigns a number 1-4 to the geodesic based on the quadrant of the boundary sphere where the geodesic terminates. If the geodesic doesn't reach the boundary sphere, it will be assigned 0.

- **Equatorial passes:** This allows the user to track the number of times a geodesic passes through the equatorial plane $\theta = \pi/2$. If the geodesic escapes to infinity, this number will be positive: if it falls into a horizon, it will get a minus sign. This is typically useful in the context of photon rings, which are indexed by the number of times they pass through the equatorial plane.
- **Closest radius:** This diagnostic keeps track of the smallest value for the radial coordinate a geodesic encounters. If the geodesic falls into a horizon, the code will report $r = 0$.
- **Equatorial emission:** This keeps track of the brightness intensity of the geodesic assuming a particular model for emission from an equatorial disc. The calculation of this brightness intensity is described in more detail in Section 3.3.

These different terminations and diagnostics allow the user to study the geometry of exotic spacetimes, and enable the analysis of their potential photon rings even though this may not be feasible analytically. The documentation provides details on how to add new diagnostics and terminations in a few simple steps.

3.2 Adaptive mesh integration

One feature that we introduce in **FOORT** is a specific adaptive mesh which allows very high image resolutions to be achieved by integrating only a relatively low number of geodesics. Our adaptive mesh algorithm starts by integrating an evenly spaced, square grid of pixels. Each pixel is then assigned a “weight”, based on how much the specified image feature changes in its nearby pixels. Based on all of these pixel weights, at each iteration the algorithm decides to divide the squares associated to pixels of the highest weights.

The tuneable parameters of the algorithm are the initial number of pixels to integrate, n_{init}^2 ; the maximum number of subdivisions that may be made, s_{max} (the initial grid is considered to be the first subdivision, so necessarily $s_{\text{max}} \geq 1$), the maximum number of pixels allowed to integrate n_{max} (with necessarily $n_{\text{init}}^2 \leq n_{\text{max}}$); and the number of pixels to subdivide each iteration, n_{iter} .

The algorithm can then be broken down into steps as follows:

Step 0. Integrate a $n_{\text{init}} \times n_{\text{init}}$ square grid of pixels.

Note that the mesh algorithm creates a grid that is already the maximum possible size, i.e. $(n_{\text{init}} - 1) * 2^{s-1} + 1$. Pixel row and column coordinates then lie between 0 and $(n_{\text{init}} - 1) * 2^{s-1}$; the initial grid is all pixels with coordinates $(i, j) \times 2^{s_{\text{max}}-1}$ for $0 \leq i, j \leq n_{\text{init}} - 1$. See Fig. 5.

Step 1. Calculate the “weight” of all pixels, i.e. a quantitative measure of how much a specified image feature changes in nearby pixels.

A particular diagnostic is set as the mesh diagnostic in the **FOORT** configuration. This mesh diagnostic assigns a *value* to each geodesic at the end of its integration; this value can be a vector of real numbers of any size (determined by the particular diagnostic). Every diagnostic also specifies a way of determining a *distance* between such values. The *weight* of a pixel is then determined as the maximum of the *distances* between itself and its lower, right, and low-right neighbors — i.e. between itself and the other three pixels that define its “square” (see Fig. 5).

For example, the diagnostic **FourColorScreen** assigns a number between 1 and 4 to each geodesic as its *value*, depending on which color has been assigned. The distance between two pixels is then the discrete metric, i.e. 0 if their colors are the same and 1 if they are different. Another example is the diagnostic **GeodesicPosition**: this assigns to a pixel as value the angular coordinates (θ, ϕ) of the position of the geodesic at termination. The distance between two pixels is then cord distance on the unit sphere between these two sets of coordinates.

- Step 2. Order all pixels by their weight and select the n_{iter} highest-weight pixels.
- Step 3. Subdivide the squares associated with all of these highest-weight pixels and integrate geodesics for these pixels.
- Step 4. Go back to step 1 as long as the total number of geodesics integrated is still smaller than n_{max} .

FOORT keeps track of all pixels and their properties in a specific **struct**. Every pixel keeps track of:

- Its position on the screen (i.e. pixel coordinates)
- Its weight
- Its subdivision level s . A subdivision level of s implies its right and lower neighbors must be located at a distance of 2^{s-1} pixels on the grid. The pixels on the rightmost column and bottom row are assigned $s = s_{max}$.
- A pointer to the pixel **struct** of its right, lower, and lower-right neighbors; these are always at a distance governed by the pixel's subdivision level s . Together with itself, these four pixels define the pixel's own “square”; see Fig. 5.
- A point to the pixel **struct** of its upper and left neighbor; these are solely defined as being *reciprocal* with respect to the right- and upper-neighbors. For example, pixel p_1 has p_2 as its *left* neighbor if p_2 has p_1 as its *right* neighbor. See Fig. 5.

Having each pixel **struct** contain pointers to its neighbor structs allows efficient traversal of the grid around any given pixel when needed. Any alternative implementation of storing pixels in a pre-ordered array or matrix array becomes computationally infeasible for large grids, especially when adding intermediate pixels in the adaptive mesh iterations.

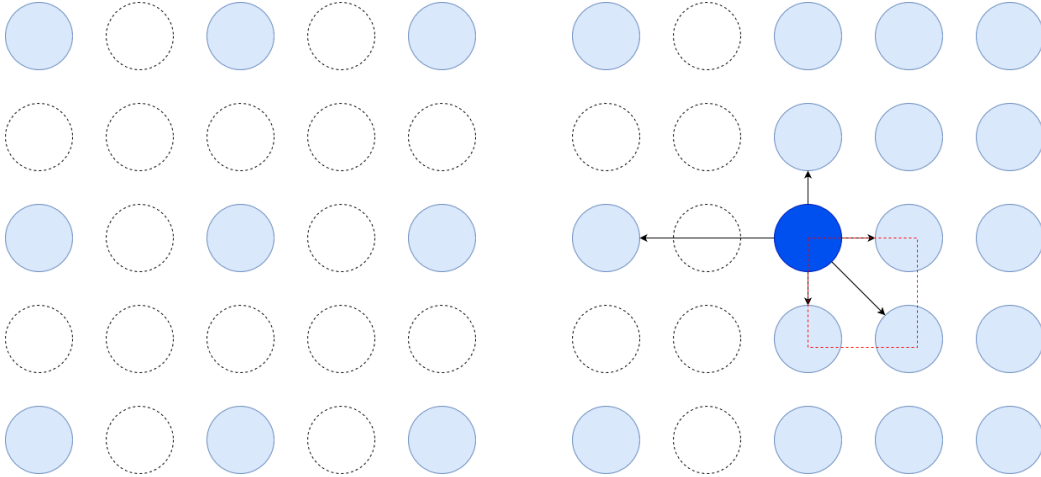


Figure 5: (*left*) The total grid generated for $n_{init} = 3$, $s_{max} = 2$ has 25 pixels total. The initial pixels integrated in Step 0 are indicated in blue. (*right*) The grid with $n_{init} = 3$, $s_{max} = 2$ after some number of integration iterations. The integrated pixels are indicated in blue. The neighbors of the dark blue central pixel are given by the arrows, and its “square” is given by the four pixels in the red dashed lines. Note that the dark blue pixel has $s = s_{max} = 2$.

3.3 Emission models and interferometry

(*DRM: — radiative transfer (disc), equatorial passes (discuss non equatorial metrics) — visamps*)

FOORT has the option to model an accretion disk in the spacetime of interest, serving as a light source whose photons are ray-traced to the observer. At the moment, the only emission model that is implemented is the **GLMJohnsonSU** model [5]. This emission profile, restricted to the equatorial plane, is given by

$$I_0(r) = ((r - \mu)^2 + \sigma^2)^{-1/2} \exp \left(-\frac{1}{2} \left[\gamma + \operatorname{arcsinh} \left(\frac{r - \mu}{\sigma} \right) \right]^2 \right), \quad (3.1)$$

where the parameters μ, σ, γ can be chosen to change the disc profile.

The total brightness intensity for the geodesic is calculated as:

$$I_{\text{tot}} = \sum_{n=1}^{N_{\text{max}}} \zeta_n I_0(p^{(n)}) g(p^{(n)})^f. \quad (3.2)$$

In this expression, N_{max} is the number of times the geodesic passes through the equatorial plane¹ and ζ_n is the geometric fudge factor used to mimic the effects of a geometrically thick disc: $\zeta_1 = 1$ and $\zeta_{n>1} = \zeta$. $I_0(p^{(n)})$ is the local intensity at the n th point of passing $p^{(n)}$, $g(p^{(n)})$ is the redshift factor and f is the power of the latter².

In order to determine the redshift factor, we need to assume a model for the fluid velocity. At the moment, the **GeneralCircularRadialFluid** model in FOORT allows for a mix of (sub)Keplerian circular orbits and radially infalling orbits. It takes three parameters: the sub-Keplerian parameter ξ , and the mixing parameters β_r, β_ϕ , all in $[0, 1]$. It is briefly summarized below, following e.g. [3, 14]. The equatorial fluid four-velocity at a point p is expanded as *(SS: make sure to cite AART and 11, 66 in AART)*

$$u = u^t \partial_t + u^r \partial_r + u^\phi \partial_\phi \quad (3.3)$$

$$\equiv u^t (\partial_t - \iota \partial_r + \Omega \partial_\phi) \quad (3.4)$$

and determined in the following steps:

1. First the ISCO, if it exists, is determined using a bisection method. The conserved quantities associated to this ISCO are determined and saved.
2. Now the 'circular velocity' vector u_{cir} is determined at the point p . The calculation depends on whether the radial coordinate of the point p is in- or outside the ISCO, as detailed in App. B, and involves optional rescaling with the sub-Keplerian parameter ξ .
3. The radial infal velocity vector u_{rad} is calculated - again detailed in App. B.3
4. The final velocity vector is now calculated as a mix of the two above using the mixing parameters β^r, β^ϕ as:

$$u^r = u_{\text{cir}}^r + (1 - \beta_r)(u_{\text{rad}}^r - u_{\text{cir}}^r), \quad (3.5)$$

$$\Omega = \Omega_{\text{cir}} + (1 - \beta_\phi)(\Omega_{\text{rad}} - \Omega_{\text{cir}}), \quad (3.6)$$

$$u^t = \sqrt{\frac{-1 - g_{rr}(u^r)^2}{g_{tt} + 2g_{t\phi}\Omega + g_{\phi\phi}\Omega^2}}, \quad (3.7)$$

$$u^\phi = u^t \Omega, \quad (3.8)$$

where (3.7) follows from the normalization $u_\mu u^\mu = -1$. It should be noted that, in general, the flow described by Eqs. (3.5) - (3.8) is not geodesic. With this model for the fluid velocity, the redshift factor can now be determined as

$$g(p^{(n)}) = \frac{-E}{u(p^{(n)})_\mu p^\mu}, \quad (3.9)$$

¹ N_{max} can also be set to a fixed value, thus avoiding higher order rings that are e.g. not fully resolved.

²This can be either 3 or 4, and the difference is *(SS: explain)*

where $u(p^{(n)})$ represents the fluid velocity at the point of passage through the disk, whereas p represents the current velocity of the geodesic. *(SS: horrible notation at the moment. u should be geodesic velocity, something else for the fluid velocity (maybe w)?)*

The resulting image of the measured intensity profile can be converted to Fourier space, in order to extract the visibility amplitudes - the actual data collected in VLBI experiments. FOORT comes with two Python-packages; pyFOORT as a general output handler with inbuilt functions for creating images, and PhotonRingInterferometry *(SS: change name?)* that contains functions to convert FOORT data into visibility amplitudes.

4 Demonstrations & Applications

— fuzzball flex, BHs

Four color, emission, visamp van Kerr, ander BH, fuzzball, BS.

Aside from analytical metrics, FOORT can in principle also be used to ray-trace numerically obtained metrics, as long as they can be read in from files. An example is the boson star (BS) metric (see Appendix A.3), which represents a solitonic BS that is sufficiently compact to have a pair of light rings. *(SS: figure)*

Acknowledgments

We especially want to thank everyone who has tested FOORT in some capacity or contributed to various features: A. Cárdenas-Avendaño (for assistance in visibility amplitudes and the Python implementation of their generation), L. Van Dael (for implementing the SUSY black hole and fuzzball metrics), P. Fernandes (for the Apple makefile, contributions to the Python plotting scripts, and testing/debugging various features related especially to Kerr-Schild coordinates), S. Giri (for the Arch makefile, and testing/debugging various features).

Additionally, we would like to thank J. Davelaar, S. Gralla, T. Hertog, L. Küchler, T. Li, A. Lupsasca, H. Olivares, H. Pagnat, B. Ripperda, B. Vercnocke for interesting discussions and earlier ray-tracing collaborations. D.R.M. is supported by FWO Research Project G.0926.17N. F.B. acknowledges support from the FED-tWIN programme (profile Prf-2020-004, project “ENERGY”) issued by BELSPO. S.S. is supported by the Centre for Doctoral Training (CDT) at the University of Cambridge funded through STFC. This work is also partially supported by the KU Leuven C1 grant ZKD1118 C16/16/005. Computing resources were provided and supported by the VSC (Flemish Supercomputer Center), funded by the Research Foundation Flanders (FWO) and the Flemish Government – department EWI.

References

- [1] F. Bacchini, B. Ripperda, A. Y. Chen, and L. Sironi. Generalized, energy-conserving numerical simulations of particles in general relativity. i. time-like and null geodesics. *The Astrophysical Journal Supplement Series*, 237(1):6, jul 2018.
- [2] Thomas Bronzwaer, Jordy Davelaar, Ziri Younsi, Monika Mościbrodzka, Heino Falcke, Michael Kramer, and Luciano Rezzolla. Raptor-i. time-dependent radiative transfer in arbitrary spacetimes. *Astronomy & Astrophysics*, 613:A2, 2018.
- [3] Alejandro Cárdenas-Avendaño, Alexandru Lupsasca, and Hengrui Zhu. Adaptive analytical ray tracing of black hole photon rings. *Phys. Rev. D*, 107:043030, Feb 2023.

- [4] Joshua C. Dolence, Charles F. Gammie, Monika Mościbrodzka, and Po Kin Leung. gr-monty: A monte carlo code for relativistic radiative transport. *The Astrophysical Journal Supplement Series*, 184(2):387, sep 2009.
- [5] Samuel E Gralla, Alexandru Lupsasca, and Daniel P Marrone. The shape of the black hole photon ring: A precise test of strong-field general relativity. *Physical Review D*, 102(12):124004, 2020.
- [6] T. Helfer, U. Sperhake, R. Croft, M. Radia, B.-X. Ge, and E. A. Lim. Malaise and remedy of binary boson-star initial data. *Class. Quant. Grav.*, 39(7):074001, 2022.
- [7] Tim Johannsen. Regular black hole metric with three constants of motion. *Physical Review D*, 88(4):044002, 2013.
- [8] Michael D Johnson, Alexandru Lupsasca, Andrew Strominger, George N Wong, Shahar Hadar, Daniel Kapec, Ramesh Narayan, Andrew Chael, Charles F Gammie, Peter Galison, et al. Universal interferometric signatures of a black hole’s photon ring. *Science advances*, 6(12):eaaz1310, 2020.
- [9] Finn Larsen. Rotating kaluza–klein black holes. *Nuclear Physics B*, 575(1-2):211–230, 2000.
- [10] Alexandru Lupsasca, Daniel R Mayerson, Bart Ripperda, and Seppe Staelens. A beginner’s guide to black hole imaging and associated tests of general relativity. In *Recent Progress on Gravity Tests: Challenges and Future Perspectives*, pages 183–237. Springer, 2024.
- [11] VS Manko and ID Novikov. Generalizations of the Kerr and Kerr-Newman metrics possessing an arbitrary set of mass-multipole moments. *Classical and Quantum Gravity*, 9(11):2477, 1992.
- [12] Thomas Müller. Einstein rings as a tool for estimating distances and the mass of a schwarzschild black hole. *Physical Review D—Particles, Fields, Gravitation, and Cosmology*, 77(12):124042, 2008.
- [13] Scott C. Noble, Po Kin Leung, Charles F. Gammie, and Laura G. Book. Simulating the emission and outflows from accretion discs. *Classical and Quantum Gravity*, 24(12):S259–S274, June 2007.
- [14] Hung-Yi Pu, Kazunori Akiyama, and Keiichi Asada. The effects of accretion flow dynamics on the black hole shadow of sagittarius a*. *The Astrophysical Journal*, 831(1):4, oct 2016.
- [15] Dean Rasheed. The Rotating dyonic black holes of Kaluza-Klein theory. *Nucl. Phys. B*, 454:379–401, 1995.
- [16] Seppe Staelens, Daniel R. Mayerson, Fabio Bacchini, Bart Ripperda, and Lorenzo Küchler. Black hole photon rings beyond general relativity. *Phys. Rev. D*, 107:124026, Jun 2023.

A Implemented metrics

FOORT comes with several implemented black hole metrics, as well as a fuzzball and boson star example. The modularity of the code makes it very straightforward to add new metrics to the code, as explained in the documentation. This appendix summarizes the spacetime metrics that are currently available.

A.1 Black holes

FOORT features a collection of black hole metrics that inherit from a parent **SphericalHorizonMetric** class, which assumes that there is an event horizon at a fixed value of the radial coordinate $r = r_H$.

A.1.1 Kerr

The Kerr metric is implemented in standard Boyer-Lindquist coordinates:

$$ds^2 = - \left(1 - \frac{2Mr}{\Sigma} \right) dt^2 + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 + \frac{(r^2 + a^2)^2 - \Delta a^2 \sin^2 \theta}{\Sigma} \sin^2 \theta d\phi^2 \quad (\text{A.1})$$

$$- \frac{4Mra \sin^2 \theta}{\Sigma} dt d\phi \quad (\text{A.2})$$

with

$$\Sigma = r^2 + a^2 \cos^2 \theta, \quad \Delta = r^2 + a^2 - 2Mr \quad (\text{A.3})$$

and $a \in [-M, M]$ the rotational parameter.

It is also implemented in the Kerr-Schild form: *(SS: to do)*

A.1.2 Johannsen

(SS: copied from photon rings paper)

The Johannsen metric is constructed as a stationary and axisymmetric generalization of the Kerr metric that keeps a Carter-like constant [7]. This ensures that the geodesic equations are still integrable, facilitating an analytic treatment. The metric is given by³

$$\begin{aligned} ds^2 = & - \tilde{\Sigma} \frac{\Delta - a^2 A_2(r)^2 \sin^2 \theta}{N(r)} dt^2 + \frac{\tilde{\Sigma}}{\Delta A_5(r)} dr^2 + \tilde{\Sigma} d\theta^2 \\ & - 2a \tilde{\Sigma} \sin^2 \theta \frac{(r^2 + a^2) A_1(r) A_2(r) - \Delta}{N(r)} dt d\phi \\ & + \tilde{\Sigma} \sin^2 \theta \frac{(r^2 + a^2)^2 A_1(r)^2 - a^2 \Delta \sin^2 \theta}{N(r)} d\phi^2, \end{aligned} \quad (\text{A.4})$$

where

$$\begin{aligned} A_1(r) &= 1 + \alpha_{13} \left(\frac{M}{r} \right)^3, & \tilde{\Sigma} &= r^2 + a^2 \cos^2 \theta + \epsilon_3 \frac{M^3}{r}, \\ A_5(r) &= 1 + \alpha_{52} \left(\frac{M}{r} \right)^2, & \Delta &= r^2 - 2Mr + a^2, \\ A_2(r) &= 1 + \alpha_{22} \left(\frac{M}{r} \right)^2, & N(r) &= [(r^2 + a^2) A_1(r) - a^2 A_2(r) \sin^2 \theta]^2. \end{aligned} \quad (\text{A.5})$$

This new metric depends on four additional parameters with respect to Kerr: $\alpha_{13}, \alpha_{22}, \alpha_{52}$, and ϵ_3 . The parameter ϵ_3 is only of interest for massive particle trajectories.

The Johannsen metric is asymptotically flat and reduces to Kerr when all deviation parameters $\alpha_{13}, \alpha_{22}, \alpha_{52}, \epsilon_3$ vanish. The event horizon of this black hole is still the same as that of the Kerr metric, $r_H = M + \sqrt{M^2 - a^2}$. Demanding that the metric is regular outside of the horizon restricts the parameters as

$$\alpha_{52} > -\frac{r_H^2}{M^2}, \quad \epsilon_3 > -\frac{r_H^3}{M^3}, \quad \alpha_{13} > -\frac{r_H^3}{M^3}, \quad (\text{A.6})$$

³We consider only the first non-zero correction in the deviation functions.

together with the requirement that

$$\alpha_{13} \neq \frac{a^2 r(r^2 + \alpha_{22} M^2) \sin^2 \theta - r^3(r^2 + a^2)}{M^3(r^2 + a^2)} \quad (\text{A.7})$$

everywhere outside the horizon.

A.1.3 Manko-Novikov

(SS: copied from photon rings paper)

The Manko-Novikov metric is constructed as a solution to the vacuum Einstein equations in four dimensions [11]; it can be thought of as deforming the Kerr black hole by multipole “bumps”. Since it is a vacuum solution, the uniqueness theorems imply it must be singular; in particular, the event horizon has a singularity (only) at the equator.

The Manko-Novikov metric can be written in Boyer-Lindquist coordinates as

$$\text{d}s^2 = -f (\text{d}t - \omega \text{d}\phi)^2 + \frac{e^{2\gamma} \rho^2}{f \Delta} \text{d}r^2 + \frac{e^{2\gamma} \rho^2}{f} \text{d}\theta^2 + \frac{\Delta \sin^2 \theta}{f} \text{d}\phi^2. \quad (\text{A.8})$$

The functions ρ, Δ are given by

$$\rho^2 = (r - M)^2 - k^2 \cos^2 \theta, \quad \Delta = r^2 - 2Mr + a^2, \quad (\text{A.9})$$

where $k = \sqrt{M^2 - a^2}$. The event horizon is at

$$r_H = M + \sqrt{M^2 - a^2}. \quad (\text{A.10})$$

The functions γ, f, ω depend on the mass and rotation parameters M, a as well as an octopole-bump parameter α_3 in a highly non-trivial way — see e.g. Appendix A of [16]. When $\alpha_3 = 0$, the metric is simply Kerr. (The generic Manko-Novikov black hole depends on an infinite series of such parameters, one for each multipole that can be added.)

A.1.4 Rasheed-Larsen

(SS: copied from photon rings paper)

The Rasheed-Larsen black hole can be obtained by a Kaluza-Klein reduction of five-dimensional Einstein gravity [15, 9]. Contrary to the Johannsen metric, this black hole solution is obtained from an actual physical theory in four dimensions, with the Lagrangian

$$L = \frac{1}{16\pi G_N} \int \text{d}^4 x \left[R - 2\partial_\mu \Phi \partial^\mu \Phi - \frac{1}{4} e^{-2\sqrt{3}\Phi} F_{\mu\nu} F^{\mu\nu} \right], \quad (\text{A.11})$$

where R is the Ricci scalar, Φ is the dilaton field and $F_{\mu\nu}$ is associated with the electromagnetic gauge field A_μ .

The Rasheed-Larsen black hole is a solution to (A.11) given by

$$\text{d}s^2 = -\frac{H_3}{\sqrt{H_1 H_2}} (\text{d}t + \mathbf{B})^2 + \sqrt{H_1 H_2} \left(\frac{\text{d}r^2}{\Delta} + \text{d}\theta^2 + \frac{\Delta}{H_3} \sin^2 \theta \text{d}\phi^2 \right). \quad (\text{A.12})$$

The functions H_1, H_2, H_3, Δ , and 1-form $\mathbf{B}$ are given by

$$H_1 = r^2 + a^2 \cos^2 \theta + r(p - 2m) + \frac{p}{p+q} \frac{(p-2m)(q-2m)}{2} - \frac{p}{2m(p+q)} \sqrt{(q^2 - 4m^2)(p^2 - 4m^2)} a \cos \theta, \quad (\text{A.13})$$

$$H_2 = r^2 + a^2 \cos^2 \theta + r(q - 2m) + \frac{q}{p+q} \frac{(p-2m)(q-2m)}{2} + \frac{q}{2m(p+q)} \sqrt{(q^2 - 4m^2)(p^2 - 4m^2)} a \cos \theta, \quad (\text{A.14})$$

$$H_3 = r^2 + a^2 \cos^2 \theta - 2mr, \quad (\text{A.15})$$

$$\Delta = r^2 + a^2 - 2mr, \quad (\text{A.16})$$

$$\mathbf{B} = \sqrt{pq} \frac{(pq + 4m^2)r - m(p-2m)(q-2m)}{2m(p+q)H_3} a \sin^2 \theta d\phi. \quad (\text{A.17})$$

The solution depends on four parameters a, m, p, q . These are related to the physical quantities of the metric: mass (M), angular momentum (J), and electric & magnetic charges (Q & P) as

$$M = \frac{p+q}{4}, \quad J = \frac{\sqrt{pq}(pq + 4m^2)}{4m(p+q)} a, \quad (\text{A.18})$$

$$Q^2 = \frac{q(q^2 - 4m^2)}{4(p+q)}, \quad P^2 = \frac{p(p^2 - 4m^2)}{4(p+q)}. \quad (\text{A.19})$$

The event horizon of the Rasheed-Larsen black hole is located at

$$r_H = m + \sqrt{m^2 - a^2}, \quad (\text{A.20})$$

which implies the bound $a^2 \leq m^2$. The other parameters need to satisfy $p, q \geq 2m$, where equality corresponds to the vanishing of the electric and/or magnetic charges. The case $p = q = 2m$ corresponds to the chargeless Kerr limit (in which the mass is given by m).

A.2 Fuzzball

I assume this is the ST3Cr

A.3 Solitonic boson star

Boson stars (BS) in general relativity are described by the action of a complex scalar field φ minimally coupled to gravity

$$S = \int \frac{\sqrt{-g}}{2} \left(\frac{R}{8\pi G} - [g^{\mu\nu} \nabla_\mu \bar{\varphi} \nabla_\nu \varphi + V(\varphi)] \right) d^4x. \quad (\text{A.21})$$

FOORT provides an example of a spherically symmetric BS where the potential takes the solitonic form

$$V(\varphi) = \mu^2 |\varphi|^2 \left(1 - 2 \frac{|\varphi|^2}{\sigma_0^2} \right)^2, \quad (\text{A.22})$$

with the solitonic parameter $\sigma_0 = 0.06$. The solution is obtained by assuming the harmonic Ansatz $\varphi(t, r) = A(r)e^{i\omega t}$, where r, t are the standard time and radial coordinates in a metric of the form

$$ds^2 = -\alpha^2 dt^2 + X^2 dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (\text{A.23})$$

The frequency ω needs to be solved for, which is done with a shooting algorithm (not included in FOORT) that solves the system of differential equations governing the BS model, as described in e.g. [6]. Given a certain value for σ_0 , an entire family of BS solutions can be

obtained parameterized by the central amplitude of the scalar field $A(0)$. For every such value $A(0)$, a discrete set of BS models can be found, characterized by the frequency ω_n and the number of zero-crossings of the radial function $A(r)$. The example files in **FOORT** correspond to a ground-state model (no zero crossings in $A(r)$) with central amplitude $A(0) = 0.044$. This specific model is an example of an ultracompact boson star, featuring a pair of light rings; one stable and one unstable. Two files are given, one containing $\Phi(r) \equiv \log(\alpha(r))$ and one containing $m(r)$ where

$$X^2(r) = \left(1 - \frac{2m(r)}{r}\right)^{-1}. \quad (\text{A.24})$$

B Expressions for the fluid velocity model

These expressions are based on ..., written down for a general metric $g_{\mu\nu}$. *(SS: do we assume p_ϕ is conserved here? I believe so. This is not true for all our metrics, but I guess we would only use this fluid model in metrics that have this symmetry anyway...)*

B.1 Circular velocity outside the ISCO

For a circular geodesic at radius r , in the equatorial plane, the geodesic equation

$$\Gamma_{\mu\nu}^r p^\mu p^\nu = 0 \quad (\text{B.1})$$

should be satisfied. This expands to

$$\Gamma_{\mu\nu}^r g^{\mu 0} g^{\nu 0} E^2 + \Gamma_{\mu\nu}^r g^{\mu 3} g^{\nu 3} L^2 - 2\Gamma_{\mu\nu}^r g^{\mu 0} g^{\nu 3} EL = 0. \quad (\text{B.2})$$

This can be straightforwardly solved for $\eta = L/E$ as a quadratic equation. We now take

$$p_0 = -E = -\left(g^{tt} + g^{\phi\phi}\eta^2 - 2g^{t\phi}\eta\right)^{-1}, \quad (\text{B.3})$$

$$p_\phi = \eta E \quad (\text{B.4})$$

nb such that $p_\mu p^\mu = -1$. This gives

$$(u_{\text{circ}})_\mu = (p_0, 0, 0, p_\phi). \quad (\text{B.5})$$

To make this sub-Keplerian, we reduce the angular momentum with $\xi \in [0, 1]$ as $p_\phi \rightarrow \xi\phi$. This then warrants a recalculation of p_t to *(SS: check this)*

B.2 Circular velocity inside the ISCO

(SS: Cite Cunningham) We extend the circular flow smoothly inside the ISCO according to Cunningham's prescription. The conserved energy and angular momentum are assumed to be those of the geodesic associated with the ISCO; this requires a non-zero component for the radial momentum such that the normalization $p_\mu p^\mu = -1$ is satisfied.

B.3 Radially infalling velocity

The fluid velocity for radially infalling particles is easily determined by relating it to a geodesic that is radially infalling from infinity, with zero initial velocity. For the latter, at infinity, we have

$$(u_{\text{rad}})_\mu = (-E, 0, 0, 0), \quad E = 1 \quad (\text{B.6})$$

At a finite radius r , the normalisation $p_\mu p^\mu = -1$ implies that

$$g^{tt} E^2 + g^{tr} E p_r + g^{rr} p_r^2 = -1, \quad (\text{B.7})$$

where we leave the possibility for a tr -component in the metric (*SS: do we actually need this in the code?*). From the above, the expression for p_r follows:

$$p_r = \frac{1}{g^{rr}} \left(g^{tr} + \sqrt{(g^{tr})^2 - g^{rr} (g^{tt} + 1)} \right) \quad (\text{B.8})$$